Chapter 13
Experiments with Random Factors
Solutions

13.1S. An experiment was performed to investigate the capability of a measurement system. Ten parts were randomly selected, and two randomly selected operators measured each part three times. The tests were made in random order, and the data are shown in Table P13.1.

Table P13.1

	Operator 1			Operator 2		
Part	Measurements			Measurements		
Number	1	2	3	1	2	3
1	50	49	50	50	48	51
2	52	52	51	51	51	51
3	53	50	50	54	52	51
4	49	51	50	48	50	51
5	48	49	48	48	49	48
6	52	50	50	52	50	50
7	51	51	51	51	50	50
8	52	50	49	53	48	50
9	50	51	50	51	48	49
10	47	46	49	46	47	48

(a) Analyze the data from this experiment.

Minitab Output

ANOVA: Measurement versus Part, Operator

```
Factor     Type Levels Values
Part     random     10     1     2     3     4     5     6     7
                           8     9    10
Operator random      2     1     2

Analysis of Variance for Measurem

Source            DF         SS         MS       F      P
Part               9     99.017     11.002   18.28  0.000
Operator           1      0.417      0.417    0.69  0.427
Part*Operator      9      5.417      0.602    0.40  0.927
Error             40     60.000      1.500
Total             59    164.850

Source           Variance Error Expected Mean Square for Each Term
                component term (using restricted model)
 1 Part           1.73333   3   (4) + 3(3) + 6(1)
 2 Operator      -0.00617   3   (4) + 3(3) + 30(2)
 3 Part*Operator -0.29938   4   (4) + 3(3)
 4 Error          1.50000       (4)
```

(b) Find point estimates of the variance components using the analysis of variance method.

$$\hat{\sigma}^2 = MS_E \qquad \hat{\sigma}^2 = 1.5$$

$$\hat{\sigma}^2_{\tau\beta} = \frac{MS_{AB} - MS_E}{n} \qquad \hat{\sigma}^2_{\tau\beta} = \frac{0.6018519 - 1.5000000}{3} < 0, \text{ assume } \hat{\sigma}^2_{\tau\beta} = 0$$

$$\hat{\sigma}^2_{\beta} = \frac{MS_B - MS_{AB}}{an} \qquad \hat{\sigma}^2_{\beta} = \frac{11.001852 - 0.6018519}{2(3)} = 1.7333$$

$$\hat{\sigma}_\tau^2 = \frac{MS_A - MS_{AB}}{bn} \qquad \hat{\sigma}_\tau^2 = \frac{0.416667 - 0.6018519}{10(3)} < 0, \text{ assume } \hat{\sigma}_\tau^2 = 0$$

All estimates agree with the *Minitab* output.

13.2S. An article by Hoof and Berman ("Statistical Analysis of Power Module Thermal Test Equipment Performance", *IEEE Transactions on Components, Hybrids, and Manufacturing Technology* Vol. 11, pp. 516-520, 1988) describes an experiment conducted to investigate the capability of measurements on thermal impedance (C°/W x 100) on a power module for an induction motor starter. There are 10 parts, three operators, and three replicates. The data are shown in Table P13.2.

Table P13.2

Part	Inspector 1			Inspector 2			Inspector 3		
Number	Test 1	Test 2	Test 3	Test 1	Test 2	Test 3	Test 1	Test 2	Test 3
1	37	38	37	41	41	40	41	42	41
2	42	41	43	42	42	42	43	42	43
3	30	31	31	31	31	31	29	30	28
4	42	43	42	43	43	43	42	42	42
5	28	30	29	29	30	29	31	29	29
6	42	42	43	45	45	45	44	46	45
7	25	26	27	28	28	30	29	27	27
8	40	40	40	43	42	42	43	43	41
9	25	25	25	27	29	28	26	26	26
10	35	34	34	35	35	34	35	34	35

(a) Analyze the data from this experiment, assuming both parts and operators are random effects.

Minitab Output

ANOVA: Impedance versus Inspector, Part

```
Factor     Type Levels Values
Inspecto random      3     1     2     3
Part     random     10     1     2     3     4     5     6     7
                        8     9    10


Analysis of Variance for Impedanc

Source           DF         SS         MS       F      P
Inspecto          2      39.27      19.63    7.28  0.005
Part              9    3935.96     437.33  162.27  0.000
Inspecto*Part    18      48.51       2.70    5.27  0.000
Error            60      30.67       0.51
Total            89    4054.40


Source           Variance Error Expected Mean Square for Each Term
                component term (using restricted model)
 1 Inspecto        0.5646   3   (4) + 3(3) + 30(1)
 2 Part           48.2926   3   (4) + 3(3) + 9(2)
 3 Inspecto*Part   0.7280   4   (4) + 3(3)
 4 Error           0.5111       (4)
```

(b) Estimate the variance components using the analysis of variance method.

$$\hat{\sigma}^2 = MS_E \qquad \hat{\sigma}^2 = 0.51$$

$$\hat{\sigma}_{\tau\beta}^2 = \frac{MS_{AB} - MS_E}{n} \qquad \hat{\sigma}_{\tau\beta}^2 = \frac{2.70 - 0.51}{3} = 0.73$$

$$\hat{\sigma}_{\beta}^2 = \frac{MS_B - MS_{AB}}{an} \quad \hat{\sigma}_{\beta}^2 = \frac{437.33 - 2.70}{3(3)} = 48.29$$
$$\hat{\sigma}_{\tau}^2 = \frac{MS_A - MS_{AB}}{bn} \quad \hat{\sigma}_{\tau}^2 = \frac{19.63 - 2.70}{10(3)} = 0.56$$

All estimates agree with the *Minitab* output.

13.3. Reconsider the data in Problem 5.11. Suppose that both factors, machines and operators, are chosen at random.

(a) Analyze the data from this experiment.

	Machine			
Operator	1	2	3	4
1	109	110	108	110
	110	115	109	108
2	110	110	111	114
	112	111	109	112
3	116	112	114	120
	114	115	119	117

The following *Minitab* output contains the analysis of variance and the variance component estimates:

Minitab Output

ANOVA: Strength versus Operator, Machine

```
Factor     Type Levels Values
Operator random      3     1     2     3
Machine  random      4     1     2     3     4

Analysis of Variance for Strength

Source               DF         SS         MS       F      P
Operator              2    160.333     80.167   10.77  0.010
Machine               3     12.458      4.153    0.56  0.662
Operator*Machine      6     44.667      7.444    1.96  0.151
Error                12     45.500      3.792
Total                23    262.958

Source              Variance Error Expected Mean Square for Each Term
                   component term (using restricted model)
 1 Operator           9.0903   3   (4) + 2(3) + 8(1)
 2 Machine           -0.5486   3   (4) + 2(3) + 6(2)
 3 Operator*Machine   1.8264   4   (4) + 2(3)
 4 Error              3.7917       (4)
```

(b) Find point estimates of the variance components using the analysis of variance method.

$$\hat{\sigma}^2 = MS_E \quad \hat{\sigma}^2 = 3.79167$$
$$\hat{\sigma}_{\tau\beta}^2 = \frac{MS_{AB} - MS_E}{n} \quad \hat{\sigma}_{\tau\beta}^2 = \frac{7.44444 - 3.79167}{2} = 1.82639$$
$$\hat{\sigma}_{\beta}^2 = \frac{MS_B - MS_{AB}}{an} \quad \hat{\sigma}_{\beta}^2 = \frac{4.15278 - 7.44444}{3(2)} < 0, \text{ assume } \hat{\sigma}_{\beta}^2 = 0$$

$$\hat{\sigma}_\tau^2 = \frac{MS_A - MS_{AB}}{bn} \quad \hat{\sigma}_\tau^2 = \frac{80.16667 - 7.44444}{4(2)} = 9.09028$$

These results agree with the *Minitab* variance component analysis.

13.4. Suppose that in Problem 5.15 the furnace positions were randomly selected, resulting in a mixed model experiment. Reanalyze the data from this experiment under this new assumption. Estimate the appropriate model components.

	Temperature (°C)		
Position	800	825	850
	570	1063	565
1	565	1080	510
	583	1043	590
	528	988	526
2	547	1026	538
	521	1004	532

The following analysis assumes a restricted model:

Minitab Output

ANOVA: Density versus Position, Temperature

```
Factor     Type Levels Values
Position random      2     1     2
Temperat  fixed      3   800   825   850

Analysis of Variance for Density

Source               DF         SS         MS       F      P
Position              1       7160       7160   16.00  0.002
Temperat              2     945342     472671 1155.52  0.001
Position*Temperat     2        818        409    0.91  0.427
Error                12       5371        448
Total                17     958691

Source               Variance Error Expected Mean Square for Each Term
                    component term (using restricted model)
 1 Position            745.83   4   (4) + 9(1)
 2 Temperat                     3   (4) + 3(3) + 6Q[2]
 3 Position*Temperat   -12.83   4   (4) + 3(3)
 4 Error               447.56       (4)
```

$$\hat{\sigma}^2 = MS_E \quad \hat{\sigma}^2 = 447.56$$

$$\hat{\sigma}_{\tau\beta}^2 = \frac{MS_{AB} - MS_E}{n} \quad \hat{\sigma}_{\tau\beta}^2 = \frac{409 - 448}{3} < 0, \text{ assume } \hat{\sigma}_{\tau\beta}^2 = 0$$

$$\hat{\sigma}_\tau^2 = \frac{MS_A - MS_E}{bn} \quad \hat{\sigma}_\tau^2 = \frac{7160 - 448}{3(3)} = 745.83$$

These results agree with the *Minitab* output.

13.5S. Reanalyze the measurement systems experiment in Problem 13.1, assuming that operators are a fixed factor. Estimate the appropriate model components.

Table P13.1

Part Number	Operator 1 Measurements 1	2	3	Operator 2 Measurements 1	2	3
1	50	49	50	50	48	51
2	52	52	51	51	51	51
3	53	50	50	54	52	51
4	49	51	50	48	50	51
5	48	49	48	48	49	48
6	52	50	50	52	50	50
7	51	51	51	51	50	50
8	52	50	49	53	48	50
9	50	51	50	51	48	49
10	47	46	49	46	47	48

The following analysis assumes a restricted model:

Minitab Output

ANOVA: Measurement versus Part, Operator

```
Factor     Type Levels Values
Part     random     10     1     2     3     4     5     6     7
                           8     9    10
Operator  fixed      2     1     2

Analysis of Variance for Measurem

Source           DF         SS         MS       F      P
Part              9     99.017     11.002    7.33  0.000
Operator          1      0.417      0.417    0.69  0.427
Part*Operator     9      5.417      0.602    0.40  0.927
Error            40     60.000      1.500
Total            59    164.850

Source           Variance Error Expected Mean Square for Each Term
                component term (using restricted model)
 1 Part            1.5836   4   (4) + 6(1)
 2 Operator                  3   (4) + 3(3) + 30Q[2]
 3 Part*Operator  -0.2994   4   (4) + 3(3)
 4 Error           1.5000       (4)
```

$$\hat{\sigma}^2 = MS_E \qquad \hat{\sigma}^2 = 1.5000$$

$$\hat{\sigma}^2_{\tau\beta} = \frac{MS_{AB} - MS_E}{n} \qquad \hat{\sigma}^2_{\tau\beta} = \frac{0.60185 - 1.5000}{3} < 0, \text{ assume } \hat{\sigma}^2_{\tau\beta} = 0$$

$$\hat{\sigma}^2_{\tau} = \frac{MS_A - MS_E}{bn} \qquad \hat{\sigma}^2_{\tau} = \frac{11.00185 - 1.50000}{2(3)} = 1.58364$$

These results agree with the *Minitab* output.

13.6. Reanalyze the measurement system experiment in Problem 13.2S, assuming that operators are a fixed factor. Estimate the appropriate model components.

Table P13.2

Part	Inspector 1			Inspector 2			Inspector 3		
Number	Test 1	Test 2	Test 3	Test 1	Test 2	Test 3	Test 1	Test 2	Test 3
1	37	38	37	41	41	40	41	42	41
2	42	41	43	42	42	42	43	42	43
3	30	31	31	31	31	31	29	30	28
4	42	43	42	43	43	43	42	42	42
5	28	30	29	29	30	29	31	29	29
6	42	42	43	45	45	45	44	46	45
7	25	26	27	28	28	30	29	27	27
8	40	40	40	43	42	42	43	43	41
9	25	25	25	27	29	28	26	26	26
10	35	34	34	35	35	34	35	34	35

Minitab Output

ANOVA: Impedance versus Inspector, Part

```
Factor     Type Levels Values
Inspecto  fixed      3     1     2     3
Part     random     10     1     2     3     4     5     6     7
                           8     9    10

Analysis of Variance for Impedanc

Source           DF         SS         MS       F      P
Inspecto          2      39.27      19.63    7.28  0.005
Part              9    3935.96     437.33  855.64  0.000
Inspecto*Part    18      48.51       2.70    5.27  0.000
Error            60      30.67       0.51
Total            89    4054.40

Source           Variance Error Expected Mean Square for Each Term
                component term (using restricted model)
 1 Inspecto                3   (4) + 3(3) + 30Q[1]
 2 Part           48.5353   4   (4) + 9(2)
 3 Inspecto*Part   0.7280   4   (4) + 3(3)
 4 Error           0.5111       (4)
```

$$\hat{\sigma}^2 = MS_E \qquad \hat{\sigma}^2 = 0.51$$

$$\hat{\sigma}^2_{\tau\beta} = \frac{MS_{AB} - MS_E}{n} \qquad \hat{\sigma}^2_{\tau\beta} = \frac{2.70 - 0.51}{3} = 0.73$$

$$\hat{\sigma}^2_{\beta} = \frac{MS_B - MS_E}{an} \qquad \hat{\sigma}^2_{\beta} = \frac{437.33 - 0.51}{3(3)} = 48.54$$

These results agree with the *Minitab* output.

13.7S. In problem 5.11, suppose that there are only four machines of interest, but the operators were selected at random.

Operator	Machine			
	1	2	3	4
1	109	110	108	110
	110	115	109	108
2	110	110	111	114
	112	111	109	112
3	116	112	114	120
	114	115	119	117

(a) What type of model is appropriate?

A mixed model is appropriate.

(b) Perform the analysis and estimate the model components.

The following analysis assumes a restricted model:

Minitab Output

ANOVA: Strength versus Operator, Machine

```
Factor     Type Levels Values
Operator random      3     1     2     3
Machine   fixed      4     1     2     3     4

Analysis of Variance for Strength

Source              DF         SS         MS       F      P
Operator             2    160.333     80.167   21.14  0.000
Machine              3     12.458      4.153    0.56  0.662
Operator*Machine     6     44.667      7.444    1.96  0.151
Error               12     45.500      3.792
Total               23    262.958

Source              Variance Error Expected Mean Square for Each Term
                   component term (using restricted model)
 1 Operator            9.547   4   (4) + 8(1)
 2 Machine                     3   (4) + 2(3) + 6Q[2]
 3 Operator*Machine    1.826   4   (4) + 2(3)
 4 Error               3.792       (4)
```

$$\hat{\sigma}^2 = MS_E \qquad \hat{\sigma}^2 = 3.792$$

$$\hat{\sigma}^2_{\tau\beta} = \frac{MS_{AB} - MS_E}{n} \qquad \hat{\sigma}^2_{\tau\beta} = \frac{7.444 - 3.792}{2} = 1.826$$

$$\hat{\sigma}^2_{\tau} = \frac{MS_A - MS_E}{bn} \qquad \hat{\sigma}^2_{\tau} = \frac{80.167 - 3.792}{4(2)} = 9.547$$

These results agree with the *Minitab* output.

13.8. Rework Problem 13.4 using the REML method.

Position	Temperature (°C)		
	800	825	850
	570	1063	565
1	565	1080	510
	583	1043	590

	528	988	526
2	547	1026	538
	521	1004	532

The JMP REML output is shown below. The variance components are similar to those calculated in Problem 13.4.

JMP Output

RSquare	0.993347
RSquare Adj	0.99246
Root Mean Square Error	21.15551
Mean of Response	709.9444
Observations (or Sum Wgts)	18

Parameter Estimates

Term	Estimate	Std Error	DFDen	t Ratio	Prob>|t|
Intercept	709.94444	19.94444	1	35.60	0.0179*
Temperature[800]	-157.6111	6.741707	2	-23.38	0.0018*
Temperature[825]	324.05556	6.741707	2	48.07	0.0004*

REML Variance Component Estimates

Random Effect	Var Ratio	Var Component	Std Error	95% Lower	95% Upper	Pct of Total
Position	1.6760179	750.11111	1126.0118	-1456.832	2957.0538	63.309
Position*Temperature	-0.028674	-12.83333	149.33585	-305.5262	279.85956	-1.083
Residual		447.55556	182.71379	230.13858	1219.556	37.774
Total		1184.8333				100.000

Covariance Matrix of Variance Component Estimates

Random Effect	Position	Position*Temperature	Residual
Position	1267902.7	-6197.276	-1.412e-9
Position*Temperature	-6197.276	22301.197	-11128.11
Residual	-1.412e-9	-11128.11	33384.329

Fixed Effect Tests

Source	Nparm	DF	DFDen	F Ratio	Prob > F
Temperature	2	2	2	1155.518	0.0009*

13.9. Rework Problem 13.5 using the REML method.

Table P13.1

Part Number	Operator 1 Measurements 1	2	3	Operator 2 Measurements 1	2	3
1	50	49	50	50	48	51
2	52	52	51	51	51	51
3	53	50	50	54	52	51
4	49	51	50	48	50	51
5	48	49	48	48	49	48
6	52	50	50	52	50	50
7	51	51	51	51	50	50
8	52	50	49	53	48	50
9	50	51	50	51	48	49
10	47	46	49	46	47	48

The JMP REML output is shown below. The variance components are similar to those calculated in Problem 13.5.

JMP Output

RSquare	0.420766
RSquare Adj	0.410779
Root Mean Square Error	1.224745
Mean of Response	49.95
Observations (or Sum Wgts)	60

Parameter Estimates

Term	Estimate	Std Error	DFDen	t Ratio	Prob>|t|
Intercept	49.95	0.42821	9	116.65	<.0001*
Operator[Operator 1]	0.0833333	0.100154	9	0.83	0.4269

REML Variance Component Estimates

Random Effect	Var Ratio	Var Component	Std Error	95% Lower	95% Upper	Pct of Total
Part Number	1.1555556	1.7333333	0.8656795	0.0366326	3.430034	59.078
Part Number*Operator	-0.199588	-0.299383	0.1464372	-0.586394	-0.012371	-10.204
Residual		1.5	0.3354102	1.0110933	2.4556912	51.126
Total		2.9339506				100.000

Covariance Matrix of Variance Component Estimates

Random Effect	Part Number	Part Number*Operator	Residual
Part Number	0.749401	-0.004472	-1.43e-14
Part Number*Operator	-0.004472	0.0214438	-0.0375
Residual	-1.43e-14	-0.0375	0.1125

Fixed Effect Tests

Source	Nparm	DF	DFDen	F Ratio	Prob > F
Operator	1	1	9	0.6923	0.4269

13.10. Rework Problem 13.6 using the REML method.

Table P13.2

Part	Inspector 1			Inspector 2			Inspector 3		
Number	Test 1	Test 2	Test 3	Test 1	Test 2	Test 3	Test 1	Test 2	Test 3
1	37	38	37	41	41	40	41	42	41
2	42	41	43	42	42	42	43	42	43
3	30	31	31	31	31	31	29	30	28
4	42	43	42	43	43	43	42	42	42
5	28	30	29	29	30	29	31	29	29
6	42	42	43	45	45	45	44	46	45
7	25	26	27	28	28	30	29	27	27
8	40	40	40	43	42	42	43	43	41
9	25	25	25	27	29	28	26	26	26
10	35	34	34	35	35	34	35	34	35

The JMP REML output is shown below. The variance components are similar to those calculated in Problem 13.6.

JMP Output

RSquare	0.992005
RSquare Adj	0.991821
Root Mean Square Error	0.71492
Mean of Response	35.8
Observations (or Sum Wgts)	90

Parameter Estimates

Term	Estimate	Std Error	DFDen	t Ratio	Prob>|t|
Intercept	35.8	2.20436	9	16.24	<.0001*
Inspector[Inspector 1]	-0.9	0.244725	18	-3.68	0.0017*
Inspector[Inspector 2]	0.6666667	0.244725	18	2.72	0.0139*

REML Variance Component Estimates

Random Effect	Var Ratio	Var Component	Std Error	95% Lower	95% Upper	Pct of Total
Part Number	94.485507	48.292593	22.906727	3.3962334	93.188952	97.498
Part Number*Inspector	1.4243156	0.7279835	0.3010625	0.1379119	1.3180552	1.470
Residual		0.5111111	0.0933157	0.3681575	0.757543	1.032
Total		49.531687				100.000

Covariance Matrix of Variance Component Estimates

Random Effect	Part Number	Part Number*Inspector	Residual
Part Number	524.71813	-0.02989	-2.76e-13
Part Number*Inspector	-0.02989	0.0906386	-0.002903
Residual	-2.76e-13	-0.002903	0.0087078

Fixed Effect Tests

Source	Nparm	DF	DFDen	F Ratio	Prob > F
Inspector	2	2	18	7.2849	0.0048*

13.11. Rework Problem 13.7 using the REML method.

Operator	Machine			
	1	2	3	4
1	109	110	108	110
	110	115	109	108
2	110	110	111	114
	112	111	109	112
3	116	112	114	120
	114	115	119	117

The JMP REML output is shown below. The variance components are similar to those calculated in Problem 13.7.

JMP Output

RSquare	0.78154
RSquare Adj	0.748771
Root Mean Square Error	1.94722
Mean of Response	112.2917
Observations (or Sum Wgts)	24

Parameter Estimates

Term	Estimate	Std Error	DFDen	t Ratio	Prob>|t|
Intercept	112.29167	1.827643	2	61.44	0.0003*
Machine[1]	-0.458333	0.964653	6	-0.48	0.6515
Machine[2]	-0.125	0.964653	6	-0.13	0.9011
Machine[3]	-0.625	0.964653	6	-0.65	0.5410

REML Variance Component Estimates

Random Effect	Var Ratio	Var Component	Std Error	95% Lower	95% Upper	Pct of Total
Operator	2.3974359	9.0902778	10.035225	-10.5784	28.758958	61.804
Operator*Machine	0.481685	1.8263889	2.2841505	-2.650464	6.3032416	12.417
Residual		3.7916667	1.5479414	1.9497217	10.332013	25.779
Total		14.708333				100.000

Covariance Matrix of Variance Component Estimates

Random Effect	Operator	Operator*Machine	Residual
Operator	100.70575	-1.154578	1.686e-12
Operator*Machine	-1.154578	5.2173434	-1.198061
Residual	1.686e-12	-1.198061	2.3961227

Fixed Effect Tests

Source	Nparm	DF	DFDen	F Ratio	Prob > F

Source	Nparm	DF	DFDen	F Ratio	Prob > F
Machine	3	3	6	0.5578	0.6619

13.12. Consider the three-factor factorial design in Example 13.4. Propose appropriate test statistics for all main effects and interactions. Repeat for the case where *A* and *B* are fixed and *C* is random.

If all three factors are random there are no exact tests on main effects. We could use the following:

$$A: F = \frac{MS_A + MS_{ABC}}{MS_{AB} + MS_{AC}}$$

$$B: F = \frac{MS_B + MS_{ABC}}{MS_{AB} + MS_{BC}}$$

$$C: F = \frac{MS_C + MS_{ABC}}{MS_{AC} + MS_{BC}}$$

If *A* and *B* are fixed and *C* is random, the expected mean squares are (assuming the restricted form of the model):

Factor	F *a* *i*	F *b* *j*	R *c* *k*	R *n* *l*	E(*MS*)
τ_i	0	b	c	n	$\sigma^2 + bn\sigma^2_{\tau\gamma} + bcn\sum \frac{\tau_i^2}{(a-1)}$
β_j	a	0	c	n	$\sigma^2 + an\sigma^2_{\beta\gamma} + acn\sum \frac{\beta_j^2}{(b-1)}$
γ_k	a	b	1	n	$\sigma^2 + abn\sigma^2_{\gamma}$
$(\tau\beta)_{ij}$	0	0	c	n	$\sigma^2 + n\sigma^2_{\tau\beta\gamma} + cn\sum\sum \frac{(\tau\beta)_{ij}^2}{(a-1)(b-1)}$
$(\tau\gamma)_{ik}$	0	b	1	n	$\sigma^2 + bn\sigma^2_{\tau\gamma}$
$(\beta\gamma)_{jk}$	a	0	1	n	$\sigma^2 + n\sigma^2_{\tau\beta\gamma}$
$(\tau\beta\gamma)_{ijk}$	0	0	1	n	$\sigma^2 + n\sigma^2_{\tau\beta\gamma}$
$\varepsilon_{(ijk)l}$	1	1	1	1	σ^2

These are exact tests for all effects.

13.13. Consider the experiment in Example 13.5. Analyze the data for the case where *A*, *B*, and *C* are random.

Minitab Output

ANOVA: Drop versus Temp, Operator, Gauge

```
Factor     Type Levels Values
Temp     random      3    60    75    90
Operator random      4     1     2     3     4
Gauge    random      3     1     2     3

Analysis of Variance for Drop
```

```
Source                 DF         SS         MS       F      P
Temp                    2    1023.36     511.68    2.30  0.171 x
Operator                3     423.82     141.27    0.63  0.616 x
Gauge                   2       7.19       3.60    0.06  0.938 x
Temp*Operator           6    1211.97     202.00   14.59  0.000
Temp*Gauge              4     137.89      34.47    2.49  0.099
Operator*Gauge          6     209.47      34.91    2.52  0.081
Temp*Operator*Gauge    12     166.11      13.84    0.65  0.788
Error                  36     770.50      21.40
Total                  71    3950.32

x Not an exact F-test.

Source                 Variance Error Expected Mean Square for Each Term
                      component term (using restricted model)
 1 Temp                  12.044   *   (8) + 2(7) + 8(5) + 6(4) + 24(1)
 2 Operator              -4.544   *   (8) + 2(7) + 6(6) + 6(4) + 18(2)
 3 Gauge                 -2.164   *   (8) + 2(7) + 6(6) + 8(5) + 24(3)
 4 Temp*Operator         31.359   7   (8) + 2(7) + 6(4)
 5 Temp*Gauge             2.579   7   (8) + 2(7) + 8(5)
 6 Operator*Gauge         3.512   7   (8) + 2(7) + 6(6)
 7 Temp*Operator*Gauge   -3.780   8   (8) + 2(7)
 8 Error                 21.403       (8)

* Synthesized Test.

Error Terms for Synthesized Tests

Source                  Error DF  Error MS  Synthesis of Error MS
 1 Temp                     6.97    222.63   (4) + (5) - (7)
 2 Operator                 7.09    223.06   (4) + (6) - (7)
 3 Gauge                    5.98     55.54   (5) + (6) - (7)
```

Since all three factors are random there are no exact tests on main effects. *Minitab* uses an approximate *F* test for the these factors.

13.14. Derive the expected mean squares shown in Table 13.10.

Factor	F *a* *i*	R *b* *j*	R *c* *k*	R *n* *l*	E(*MS*)
τ_i	0	b	c	n	$\sigma^2 + n\sigma^2_{\tau\beta\gamma} + bn\sigma^2_{\tau\gamma} + cn\sigma^2_{\tau\beta} + bcn\sum \frac{\tau_i^2}{(a-1)}$
β_j	a	1	c	n	$\sigma^2 + an\sigma^2_{\beta\gamma} + acn\sigma^2_{\beta}$
γ_k	a	b	1	n	$\sigma^2 + an\sigma^2_{\beta\gamma} + abn\sigma^2_{\gamma}$
$(\tau\beta)_{ij}$	0	1	c	n	$\sigma^2 + n\sigma^2_{\tau\beta\gamma} + cn\sigma^2_{\tau\beta}$
$(\tau\gamma)_{ik}$	0	b	1	n	$\sigma^2 + n\sigma^2_{\tau\beta\gamma} + bn\sigma^2_{\tau\gamma}$
$(\beta\gamma)_{jk}$	a	1	1	n	$\sigma^2 + an\sigma^2_{\beta\gamma}$
$(\tau\beta\gamma)_{ijk}$	0	1	1	n	$\sigma^2 + n\sigma^2_{\tau\beta\gamma}$
ε_{ijkl}	1	1	1	1	σ^2

13.15S. In Problem 5.21, assume that the three operators were selected at random. Analyze the data under these conditions and draw conclusions. Estimate the variance components.

Minitab Output

ANOVA: Score versus Cycle Time, Operator, Temperature

```
Factor     Type Levels Values
Cycle Ti   fixed      3    40    50    60
Operator random       3     1     2     3
Temperat   fixed      2   300   350

Analysis of Variance for Score

Source                        DF         SS         MS       F      P
Cycle Ti                       2    436.000    218.000    2.45  0.202
Operator                       2    261.333    130.667   39.86  0.000
Temperat                       1     50.074     50.074    8.89  0.096
Cycle Ti*Operator              4    355.667     88.917   27.13  0.000
Cycle Ti*Temperat              2     78.815     39.407    3.41  0.137
Operator*Temperat              2     11.259      5.630    1.72  0.194
Cycle Ti*Operator*Temperat     4     46.185     11.546    3.52  0.016
Error                         36    118.000      3.278
Total                         53   1357.333

Source                         Variance Error Expected Mean Square for Each Term
                              component term (using restricted model)
 1 Cycle Ti                              4   (8) + 6(4) + 18Q[1]
 2 Operator                       7.0772   8   (8) + 18(2)
 3 Temperat                              6   (8) + 9(6) + 27Q[3]
 4 Cycle Ti*Operator             14.2731   8   (8) + 6(4)
 5 Cycle Ti*Temperat                     7   (8) + 3(7) + 9Q[5]
 6 Operator*Temperat              0.2613   8   (8) + 9(6)
 7 Cycle Ti*Operator*Temperat     2.7562   8   (8) + 3(7)
 8 Error                          3.2778       (8)
```

The following calculations agree with the *Minitab* results:

$$\hat{\sigma}^2 = MS_E \qquad \hat{\sigma}^2 = 3.27778$$

$$\hat{\sigma}^2_{\tau\beta\gamma} = \frac{MS_{ABC} - MS_E}{n} \qquad \hat{\sigma}^2_{\tau\beta\gamma} = \frac{11.546296 - 3.277778}{3} = 2.7562$$

$$\hat{\sigma}^2_{\tau\beta} = \frac{MS_{AB} - MS_E}{cn} \qquad \hat{\sigma}^2_{\tau\beta} = \frac{88.91667 - 3.277778}{2(3)} = 14.27315$$

$$\hat{\sigma}^2_{\beta\gamma} = \frac{MS_{BC} - MS_E}{an} \qquad \hat{\sigma}^2_{\tau\gamma} = \frac{5.629630 - 3.277778}{3(3)} = 0.26132$$

$$\hat{\sigma}^2_{\beta} = \frac{MS_{B} - MS_E}{acn} \qquad \hat{\sigma}^2_{\gamma} = \frac{130.66667 - 3.277778}{3(2)(3)} = 7.07716$$

13.16. Consider the three-factor factorial model

$$y_{ijk} = \mu + \tau_i + \beta_j + \gamma_k + (\tau\beta)_{ij} + (\beta\gamma)_{jk} + \varepsilon_{ijk}$$

Assuming that all the factors are random, develop the analysis of variance table, including the expected mean squares. Propose appropriate test statistics for all effects.

Source	DF	E(MS)
A	*a*-1	$\sigma^2 + c\sigma^2_{\tau\beta} + bc\sigma^2_{\tau}$
B	*b*-1	$\sigma^2 + c\sigma^2_{\tau\beta} + a\sigma^2_{\beta\gamma} + ac\sigma^2_{\beta}$
C	*c*-1	$\sigma^2 + a\sigma^2_{\beta\gamma} + ab\sigma^2_{\gamma}$
AB	(*a*-1)(*b*-1)	$\sigma^2 + c\sigma^2_{\tau\beta}$
BC	(*b*-1)(*c*-1)	$\sigma^2 + a\sigma^2_{\beta\gamma}$
Error (*AC* + *ABC*)	*b*(*a*-1)(*c*-1)	σ^2
Total	*abc*-1	

There are exact tests for all effects except *B*. To test *B*, use the statistic $F = \dfrac{MS_B + MS_E}{MS_{AB} + MS_{BC}}$

13.17. In the two-factor mixed model analysis of variance, show that $\mathrm{Cov}\left[(\tau\beta)_{ij}, (\tau\beta)_{i'j}\right] = -(1/a)^2_{\tau\beta\sigma}$ for $i \neq i'$.

Since $\sum_{i=1}^{a}(\tau\beta)_{ij} = 0$(constant) we have $V\left[\sum_{i=1}^{a}(\tau\beta)_{ij}\right] = 0$, which implies that

$$\sum_{i=1}^{a} V(\tau\beta)_{ij} + 2\binom{a}{2} Cov\left[(\tau\beta)_{ij}, (\tau\beta)_{i'j}\right] = 0$$

$$a\left[\frac{a-1}{a}\right]\sigma^2_{\tau\beta} + \frac{a!}{2!(a-2)!}(2)Cov\left[(\tau\beta)_{ij}, (\tau\beta)_{i'j}\right] = 0$$

$$(a-1)\sigma^2_{\tau\beta} + a(a-1)Cov\left[(\tau\beta)_{ij}, \tau(\beta)_{i'j}\right] = 0$$

$$Cov\left[\tau(\beta)_{ij}, (\tau\beta)_{i'j}\right] = -\left(\frac{1}{a}\right)\sigma^2_{\tau\beta}$$

13.18. Show that the method of analysis of variance always produces unbiased point estimates of the variance component in any random or mixed model.

Let $\mathbf{g}$ be the vector of mean squares from the analysis of variance, chosen so that $E(\mathbf{g})$ does not contain any fixed effects. Let $\boldsymbol{\sigma}^2$ be the vector of variance components such that $E(\mathbf{g}) = \mathbf{A}\boldsymbol{\sigma}^2$, where $\mathbf{A}$ is a matrix of constants. Now in the analysis of variance method of variance component estimation, we equate observed and expected mean squares, i.e.

$$\mathbf{g} = \mathbf{A}\mathbf{s}^2 \Rightarrow \hat{\mathbf{s}}^2 = \mathbf{A}^{-1}\mathbf{g}$$

Since $\mathbf{A}^{-1}$ always exists then,

$$E(\mathbf{s}^2) = E(\mathbf{A}^{-1})\mathbf{g} = \mathbf{A}^{-1}E(\mathbf{g}) = \mathbf{A}^{-1}(\mathbf{A}\mathbf{s}^2) = \mathbf{s}^2$$

Thus $\hat{\boldsymbol{\sigma}}^2$ is an unbiased estimator of $\boldsymbol{\sigma}^2$. This and other properties of the analysis of variance method are discussed by Searle (1971a).

13.19. Analyze the data in Problem 13.1, assuming that the operators are fixed, using both the unrestricted and restricted forms of the mixed models. Compare the results obtained from the two models.

The restricted model is as follows:

Minitab Output

ANOVA: Measurement versus Part, Operator

```
Factor     Type Levels Values
Part     random     10      1     2     3     4     5     6     7
                            8     9    10
Operator  fixed      2      1     2

Analysis of Variance for Measurem

Source           DF         SS         MS       F      P
Part              9     99.017     11.002    7.33  0.000
Operator          1      0.417      0.417    0.69  0.427
Part*Operator     9      5.417      0.602    0.40  0.927
Error            40     60.000      1.500
Total            59    164.850

Source           Variance Error Expected Mean Square for Each Term
                component term (using restricted model)
 1 Part            1.5836   4   (4) + 6(1)
 2 Operator                 3   (4) + 3(3) + 30Q[2]
 3 Part*Operator  -0.2994   4   (4) + 3(3)
 4 Error           1.5000       (4)
```

The second approach is the unrestricted mixed model.

Minitab Output

ANOVA: Measurement versus Part, Operator

```
Factor     Type Levels Values
Part     random     10      1     2     3     4     5     6     7
                            8     9    10
Operator  fixed      2      1     2

Analysis of Variance for Measurem

Source           DF         SS         MS       F      P
Part              9     99.017     11.002   18.28  0.000
Operator          1      0.417      0.417    0.69  0.427
Part*Operator     9      5.417      0.602    0.40  0.927
Error            40     60.000      1.500
Total            59    164.850

Source           Variance Error Expected Mean Square for Each Term
                component term (using unrestricted model)
 1 Part            1.7333   3   (4) + 3(3) + 6(1)
 2 Operator                 3   (4) + 3(3) + Q[2]
 3 Part*Operator  -0.2994   4   (4) + 3(3)
 4 Error           1.5000       (4)
```

Source	Sum of Squares	DF	Mean Square	E(MS)	*F*-test	*F*
A	99.016667	*a*-1=9	11.00185	$\sigma^2 + n\sigma^2_{\tau\beta} + bn\sigma^2_{\tau}$	$F = \frac{MS_A}{MS_{AB}}$	18.28
B	0.416667	*b*-1=1	0.416667	$\sigma^2 + n\sigma^2_{\tau\beta} + an\frac{\sum_{i=1}^{b}\beta_i^2}{b-1}$	$F = \frac{MS_B}{MS_{AB}}$	0.692
AB	5.416667	(*a*-1)(*b*-1)=9	0.60185	$\sigma^2 + n\sigma^2_{\tau\beta}$	$F = \frac{MS_{AB}}{MS_E}$	0.401

Error	60.000000	40	1.50000	σ^2
Total	164.85000	*nabc*-1=59		

In the unrestricted model, the *F*-test for *A* is different. The *F*-test for *A* in the unrestricted model should generally be more conservative, since MS_{AB} will generally be larger than MS_E. However, this is not the case with this particular experiment.

13.20. Consider the two-factor mixed model. Show that the standard error of the fixed factor mean (e.g. *A*) is $\left[\frac{MS_{AB}}{bn}\right]^{1/2}$.

The standard error is often used in Duncan's Multiple Range test. Duncan's Multiple Range Test requires the variance of the difference in two means, say

$$V\left(\bar{y}_{i..} - \bar{y}_{m..}\right)$$

where rows are fixed and columns are random. Now, assuming all model parameters to be independent, we have the following:

$$\left(\bar{y}_{i..} - \bar{y}_{m..}\right) = \tau_i - \tau_m + \frac{1}{b}\sum_{j=1}^{b}(\tau\beta)_{ij} - \frac{1}{b}\sum_{j=1}^{b}(\tau\beta)_{mj} + \frac{1}{bn}\sum_{j=1}^{b}\sum_{k=1}^{n}\varepsilon_{ijk} - \frac{1}{bn}\sum_{j=1}^{b}\sum_{k=1}^{n}\varepsilon_{mjk}$$

and

$$V\left(\bar{y}_{i..} - \bar{y}_{m..}\right) = \left(\frac{1}{b}\right)^2 b\sigma^2_{\tau\beta} + \left(\frac{1}{b}\right)^2 b\sigma^2_{\tau\beta} + \left(\frac{1}{bn}\right)^2 bn\sigma^2 + \left(\frac{1}{bn}\right)^2 bn\sigma^2 = \frac{2\left(\sigma^2 + n\sigma^2_{\tau\beta}\right)}{bn}$$

Since MS_{AB} estimates $\sigma^2 + n\sigma^2_{\tau\beta}$, we would use

$$\frac{2MS_{AB}}{bn}$$

as the standard error to test the difference. However, the table of ranges for Duncan's Multiple Range test already includes the constant 2.

13.21. Consider the variance components in the random model from Problem 13.1.

(a) Find an exact 95 percent confidence interval on σ^2.

$$\frac{f_E MS_E}{\chi^2_{\alpha/2,f_E}} \le \sigma^2 \le \frac{f_E MS_E}{\chi^2_{1-\alpha/2,f_E}}$$

$$\frac{(40)(1.5)}{59.34} \le \sigma^2 \le \frac{(40)(1.5)}{24.43}$$

$$1.011 \le \sigma^2 \le 2.456$$

(b) Find approximate 95 percent confidence intervals on the other variance components using the Satterthwaite method.

$\hat{\sigma}^2_{\tau\beta}$ and $\hat{\sigma}^2_{\tau}$ are negative, and the Satterthwaithe method does not apply. The confidence interval on $\hat{\sigma}^2_{\beta}$ is

$$\hat{\sigma}^2_{\beta} = \frac{MS_B - MS_{AB}}{an} \qquad \hat{\sigma}^2_{\beta} = \frac{11.001852 - 0.6018519}{2(3)} = 1.7333$$

$$r = \frac{\left(MS_B - MS_{AB}\right)^2}{\frac{MS_B^2}{(b-1)} + \frac{MS_{AB}^2}{(a-1)(b-1)}} = \frac{\left(11.001852 - 0.6018519\right)^2}{\frac{11.001852^2}{(9)} + \frac{0.6018519^2}{(1)(9)}} = 8.01826$$

$$\frac{r\hat{\sigma}^2_O}{\chi^2_{\alpha/2,r}} \le \sigma^2_{\beta} \le \frac{r\hat{\sigma}^2_{\beta}}{\chi^2_{1-\alpha/2,r}}$$

$$\frac{(8.01826)(1.7333)}{17.55752} \le \sigma^2_{\beta} \le \frac{(8.01826)(1.7333)}{2.18950}$$

$$0.79157 \le \sigma^2_{\beta} \le 6.34759$$

13.22. Use the experiment described in Problem 5.11 and assume that both factors are random. Find an exact 95 percent confidence interval on σ^2. Construct approximate 95 percent confidence interval on the other variance components using the Satterthwaite method.

$$\hat{\sigma}^2 = MS_E \qquad \hat{\sigma}^2 = 3.79167$$

$$\frac{f_E MS_E}{\chi^2_{\alpha/2,f_E}} \le \sigma^2 \le \frac{f_E MS_E}{\chi^2_{1-\alpha/2,f_E}}$$

$$\frac{(12)(3.79167)}{23.34} \le \sigma^2 \le \frac{(12)(3.79167)}{4.40}$$

$$1.9494 \le \sigma^2 \le 10.3409$$

Satterthwaite Method:

$$\hat{\sigma}^2_{\tau\beta} = \frac{MS_{AB} - MS_E}{n} \qquad \hat{\sigma}^2_{\tau\beta} = \frac{7.44444 - 3.79167}{2} = 1.82639$$

$$r = \frac{\left(MS_{AB} - MS_E\right)^2}{\frac{MS_{AB}^2}{(a-1)(b-1)} + \frac{MS_E^2}{df_E}} = \frac{\left(7.44444 - 3.79167\right)^2}{\frac{7.44444^2}{(2)(3)} + \frac{3.79167^2}{(12)}} = 1.27869$$

$$\frac{r\hat{\sigma}^2_{\beta}}{\chi^2_{\alpha/2,r}} \le \sigma^2_{\beta} \le \frac{r\sigma^2_{\beta}}{\chi^2_{1-\alpha/2,r}}$$

$\chi^2_{\alpha/2,r}$ and $\chi^2_{1-\alpha/2,r}$ were estimated by extrapolating 1.27869 between degrees of freedom of one and two in *Microsoft Excel.* More precise methods can be used as well.

$$\frac{(1.27869)(1.82639)}{5.67991} \le \sigma^2_\beta \le \frac{(1.27869)(1.82639)}{0.014820}$$

$$0.41117 \le \sigma^2_\beta \le 158.58172$$

$\hat{\sigma}^2_\beta < 0$, this variance component does not have a confidence interval using Satterthwaite's Method.

$$\hat{\sigma}^2_\tau = \frac{MS_A - MS_{AB}}{bn} \quad \hat{\sigma}^2_\tau = \frac{80.16667 - 7.44444}{4(2)} = 9.09028$$

$$r = \frac{(MS_A - MS_{AB})^2}{\frac{MS_A^2}{(a-1)} + \frac{MS_{AB}^2}{(a-1)(b-1)}} = \frac{(80.16667 - 7.44444)^2}{\frac{80.16667^2}{(2)} + \frac{7.44444^2}{(2)(3)}} = 1.64108$$

$$\frac{r\hat{\sigma}^2_\tau}{\chi^2_{\alpha/2,r}} \le \sigma^2_\tau \le \frac{r\sigma^2_\tau}{\chi^2_{1-\alpha/2,r}}$$

$$\frac{(1.64108)(9.09028)}{6.53293} \le \sigma^2_\tau \le \frac{(1.64108)(9.09028)}{0.03281}$$

$\chi^2_{\alpha/2,r}$ and $\chi^2_{1-\alpha/2,r}$ were estimated by extrapolating 1.64108 between degrees of freedom of one and two in *Microsoft Excel.* More precise methods can be used as well.

$$2.28349 \le \sigma^2_\tau \le 454.61891$$

13.23. Consider the three-factor experiment in Problem 5.21 and assume that operators were selected at random. Find an approximate 95 percent confidence interval on the operator variance component.

$$\hat{\sigma}^2_\beta = \frac{MS_B - MS_E}{acn} \quad \hat{\sigma}^2_\beta = \frac{130.66667 - 3.277778}{2(3)(3)} = 7.07716$$

$$r = \frac{(MS_B - MS_E)^2}{\frac{MS_B^2}{(b-1)} + \frac{MS_E^2}{df_E}} = \frac{(130.66667 - 3.27778)^2}{\frac{130.66667^2}{(2)} + \frac{3.27778^2}{(36)}} = 1.90085$$

$$\frac{r\hat{\sigma}^2_\beta}{\chi^2_{\alpha/2,r}} \le \sigma^2_\beta \le \frac{r\sigma^2_\beta}{\chi^2_{1-\alpha/2,r}}$$

$\chi^2_{\alpha/2,r}$ and $\chi^2_{1-\alpha/2,r}$ were estimated by extrapolating 1.90085 between degrees of freedom of one and two in *Microsoft Excel.* More precise methods can be used as well.

$$\frac{(1.90085)(7.07716)}{7.14439} \le \sigma_\beta^2 \le \frac{(1.90085)(7.07716)}{0.04571}$$

$$1.88296 \le \sigma_\beta^2 \le 294.28720$$

13.24S. Rework Problem 13.19 using the modified large-sample approach described in Section 13.6.2. Compare the two sets of confidence intervals obtained and discuss.

$$\hat{\sigma}_O^2 = \sigma_\beta^2 = \frac{MS_B - MS_{AB}}{an} \qquad \hat{\sigma}_O^2 = \frac{11.001852 - 0.6018519}{2(3)} = 1.7333$$

$$G_1 = 1 - \frac{1}{F_{0.05,9,\infty}} = 1 - \frac{1}{1.88} = 0.46809$$

$$H_1 = \frac{1}{F_{.95,9_i,\infty}} - 1 = \frac{1}{\frac{\chi^2_{.95,9}}{9}} - 1 = \frac{1}{0.370} - 1 = 1.7027$$

$$G_{ij} = \frac{\left(F_{\alpha,f_i,f_j} - 1\right)^2 - G_1^2 F_{\alpha,f_i,f_j} - H_1^2}{F_{\alpha,f_i,f_j}} = \frac{(3.18-1)^2 - (0.46809)^2(3.18) - 1.7027^2}{3.18} = 0.36366$$

$$V_L = G_1^2 c_1^2 MS_B^2 + H_1^2 c_2^2 MS_{AB}^2 + G_{11} c_1 c_2 MS_B MS_{AB}$$

$$V_L = (0.46809)^2\left(\frac{1}{6}\right)^2(11.00185)^2 + (1.7027)^2\left(\frac{1}{6}\right)^2(0.60185)^2 + (0.36366)\left(\frac{1}{6}\right)\left(\frac{1}{6}\right)(11.00185)(0.60185)$$

$$V_L = 0.83275$$

$$L = \hat{\sigma}_\beta^2 - \sqrt{V_L} = 1.7333 - \sqrt{0.83275} = 0.82075$$

13.25. Rework Problem 13.21 using the modified large-sample method described in Section 13.6.2. Compare this confidence interval with the one obtained previously and discuss.

$$\hat{\sigma}_\gamma^2 = \frac{MS_C - MS_E}{abn} \qquad \hat{\sigma}_\gamma^2 = \frac{130.66667 - 3.277778}{2(3)(3)} = 7.07716$$

$$G_1 = 1 - \frac{1}{F_{0.05,3,\infty}} = 1 - \frac{1}{2.60} = 0.61538$$

$$H_1 = \frac{1}{F_{.95,36,\infty}} - 1 = \frac{1}{\frac{\chi^2_{.95,36}}{36}} - 1 = . \frac{1}{0.64728} - 1 = 0.54493$$

$$G_{ij} = \frac{\left(F_{\alpha,f_i,f_j} - 1\right)^2 - G_1^2 F_{\alpha,f_i,f_j} - H_1^2}{F_{\alpha,f_i,f_j}} = \frac{(2.88-1)^2 - (0.61538)^2(2.88) - 0.54493^2}{2.88} = 0.74542$$

$$V_L = G_1^2 c_1^2 MS_B^2 + H_1^2 c_2^2 MS_{AB}^2 + G_{11} c_1 c_2 MS_B MS_{AB}$$

$$V_L = (0.61538)^2\left(\frac{1}{18}\right)^2(130.66667)^2 + (0.54493)^2\left(\frac{1}{18}\right)^2(3.27778)^2 + (0.74542)\left(\frac{1}{18}\right)\left(\frac{1}{18}\right)(130.66667)(3.27778)$$

$$V_L = 20.95112$$

$$L = \hat{\sigma}_\gamma^2 - \sqrt{V_L} = 7.07716 - \sqrt{20.95112} = 2.49992$$

13.26. Consider the experiment described in Problem 5.11. Estimate the variance component using the REML method. Compare the CIs to the approximate CIs found in Problem 13.22.

The JMP REML analysis below was performed with both factors, Operator and Machine, as random.

The CIs for the error variance are similar to those found in Problem 13.22. The upper CIs for the other variance components are much larger than those estimated in the JMP REML output below.

JMP Output

RSquare	0.742044
RSquare Adj	0.742044
Root Mean Square Error	1.94722
Mean of Response	112.2917
Observations (or Sum Wgts)	24

Parameter Estimates

Term	Estimate	Std Error	DFDen	t Ratio	Prob>|t|
Intercept	112.29167	1.789728	1.831	62.74	0.0005*

REML Variance Component Estimates

Random Effect	Var Ratio	Var Component	Std Error	95% Lower	95% Upper	Pct of Total
Operator	2.3974359	9.0902778	10.035225	-10.5784	28.758958	64.198
Machine	-0.144689	-0.548611	0.9124188	-2.336919	1.239697	-3.874
Operator*Machine	0.481685	1.8263889	2.2841505	-2.650464	6.3032416	12.898
Residual		3.7916667	1.5479414	1.9497217	10.332013	26.778
Total		14.159722				100.000

Covariance Matrix of Variance Component Estimates

Random Effect	Operator	Machine	Operator*Machine	Residual
Operator	100.70575	0.3848594	-1.154578	2.151e-12
Machine	0.3848594	0.8325081	-1.539438	-1.17e-13
Operator*Machine	-1.154578	-1.539438	5.2173434	-1.198061
Residual	2.151e-12	-1.17e-13	-1.198061	2.3961227

13.27S. Consider the experiment described in Problem 13.1. Analyze the data using REML. Compare the CIs to those obtained in Problem 13.21.

The JMP REML analysis below was performed with both factors, Part Number and Operator, as random.

The CIs for the Operator and Part Number Operator interaction were not calculated in Problem 13.21 due to negative estimates for the corresponding variance components. The error variance estimates and CIs found in the JMP REML output below are the same as those calculated in Problem 13.21. The upper CI for the Part Number variance estimated with the Satterthwaite method in Problem 13.21 is approximately twice the value estimated in the JMP REML analysis.

JMP Output

RSquare	0.388009
RSquare Adj	0.388009
Root Mean Square Error	1.224745
Mean of Response	49.95
Observations (or Sum Wgts)	60

Parameter Estimates

Term	Estimate	Std Error	DFDen	t Ratio	Prob>|t|
Intercept	49.95	0.424591	8.563	117.64	<.0001*

REML Variance Component Estimates

Random Effect	Var Ratio	Var Component	Std Error	95% Lower	95% Upper	Pct of Total
Part Number	1.1555556	1.7333333	0.8656795	0.0366326	3.430034	59.203
Operator	-0.004115	-0.006173	0.0218	-0.0489	0.0365544	-0.211

Random Effect	Var Ratio	Var Component	Std Error	95% Lower	95% Upper	Pct of Total
Part Number*Operator	-0.199588	-0.299383	0.1464372	-0.586394	-0.012371	-10.226
Residual		1.5	0.3354102	1.0110933	2.4556912	51.233
Total		2.9277778				100.000

Covariance Matrix of Variance Component Estimates

Random Effect	Part Number	Operator	Part Number*Operator	Residual
Part Number	0.749401	0.0004472	-0.004472	9.256e-14
Operator	0.0004472	0.0004752	-0.000894	-3.32e-16
Part Number*Operator	-0.004472	-0.000894	0.0214438	-0.0375
Residual	9.256e-14	-3.32e-16	-0.0375	0.1125

13.28. Rework Problem 13.24 using REML. Compare all sets of CIs for the variance components.

JMP uses the unrestricted approach for estimating variance components with mixed models. The JMP REML output is shown below.

The lower confidence interval on σ_β^2 (Operator) comparisons between the Satterthwaite method in Problem 13.21, the modified large sample approach in Problem 13.30, and the JMP REML output below are:

Satterthwaite $0.79157 \le \sigma_\beta^2 \le 6.34759$

Modified $0.82075 \le \sigma_\beta^2$

REML $0.03663 \le \sigma_\beta^2 \le 3.43003$

JMP Output

RSquare	0.388009
RSquare Adj	0.388009
Root Mean Square Error	1.224745
Mean of Response	49.95
Observations (or Sum Wgts)	60

Parameter Estimates

Term	Estimate	Std Error	DFDen	t Ratio	Prob>|t|
Intercept	49.95	0.424591	8.563	117.64	<.0001*

REML Variance Component Estimates

Random Effect	Var Ratio	Var Component	Std Error	95% Lower	95% Upper	Pct of Total
Part Number	1.1555556	1.7333333	0.8656795	0.0366326	3.430034	59.203
Operator	-0.004115	-0.006173	0.0218	-0.0489	0.0365544	-0.211
Part Number*Operator	-0.199588	-0.299383	0.1464372	-0.586394	-0.012371	-10.226
Residual		1.5	0.3354102	1.0110933	2.4556912	51.233
Total		2.9277778				100.000

Covariance Matrix of Variance Component Estimates

Random Effect	Part Number	Operator	Part Number*Operator	Residual
Part Number	0.749401	0.0004472	-0.004472	9.256e-14
Operator	0.0004472	0.0004752	-0.000894	-3.32e-16
Part Number*Operator	-0.004472	-0.000894	0.0214438	-0.0375
Residual	9.256e-14	-3.32e-16	-0.0375	0.1125

13.29. The levels of a random factor are sampled from a large population of possible levels.

True False

13.30. The REML method is preferred as a technique for estimating variance components because it can also provide the standard error of the variance component leading to a confidence interval.

True False

13.31. Both the ANOVA method and REML will produce the same point estimate of a variance component if the design is balanced.

True False

13.32. Random factors are always categorical.

True **False**

13.33. In a mixed model the statistical tests on fixed factors are always exactly the same as they are in the fixed effects model.

True **False**

13.34. A negative point estimate of a variance component implies that the variance of that fixed effect is zero.

True **False**

13.35. In a two-factor factorial experiment with at least one random factor

(a) the statistical tests on main effects are identical to the corresponding main effects tests in the fixed-effects case.

(b) the statistical tests on main effects are different from the corresponding main effects tests in the fixed-effects case.

(c) no test on the two-factor interaction may be made.

(d) none of the above (a-c) are true

(e) all of the above (a-c) are true

13.36. In a two-factor factorial experiment with at least one random factor

(a) the statistical test on interaction effect is identical to the corresponding interaction effect test in the fixed-effects case.

(b) the statistical test on interaction effect is different from the corresponding interaction effect test in the fixed-effects case.

(c) no test on the two factor interaction may be made.

(d) none of the above (a-c) are true

(e) all of the above (a-c) are true